

Task 1: Student Registration & Data Storage

Design a Student Registration Form using HTML5 and CSS3 that collects: Name, Email, DOB, Department, Phone Store the data in a database table and retrieve it using SELECT.

AIM:

To design a Student Registration Form using HTML5 and CSS3 that collects student details - Name, Email, Date of Birth, Department, and Phone Number and to store the submitted data into a relational database table and retrieve it using SELECT statements.

PROCEDURE:

Step 1: Create an HTML5 file with a<form>element containing fields: Name, Email, DOB, Department, Phone.

Step 2: Apply HTML5 validation attributes: required,type = "email",type = "date",patternfor phone.

Step 3: Style the form using CSS3: useflexbox/gridlayout, apply colours, borders, hover effects, and a submit button.

Step 4: Create a MySQL/SQLite database and switch to it usingCREATE DATABASEandUSE.

Step 5: Create astudentstable with appropriate data types for each field.

Step 6: Insert sample student records usingINSERT INTOstatements.

Step 7: Retrieve all records usingSELECT * FROM students;and verify the output.

Step 8: Apply optional filters usingWHERE,ORDER BY, andLIMITclauses.

PROGRAM:

Student registration.HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Student Registration</title>
<style>
body {
  font-family: Arial;
  background: #eef2f7;
  display: flex;
  justify-content: center;
  align-items: center;
  height: 100vh;
}
.container {
  background: #fff;
  padding: 20px;
  width: 350px;
  border-radius: 10px;
  box-shadow: 0 5px 15px rgba(0,0,0,0.2);
}
h2 { text-align: center; }
input, select, button {
  width: 100%;
  margin: 8px 0;
  padding: 8px;
}
button {
  background: #1a73e8;
```

```
color: white;
border: none;
cursor: pointer;
}
table {
width: 100%;
margin-top: 10px;
border-collapse: collapse;
}
td, th {
border: 1px solid #ccc;
padding: 5px;
}
</style>
</head>
<body>
<div class="container">
<h2>Student Registration</h2>
<form id="form">
  <input type="text" id="name" placeholder="Full Name" required>
  <input type="email" id="email" placeholder="Email" required>
  <input type="date" id="dob" required>
  <input type="tel" id="phone" placeholder="Phone (10 digits)" required>
  <select id="dept" required>
    <option value="">Select Department</option>
    <option>Computer Science</option>
    <option>IT</option>
    <option>ECE</option>
  </select>
  <button type="submit">Register</button>
  <button type="reset">Clear</button>
</form>
```

```
<p id="msg" style="color:green;"></p>
<table id="table" style="display:none;">
<thead><tr><th>Field</th><th>Value</th></tr></thead>
<tbody id="data"></tbody>
</table>
</div>
<script>
const form = document.getElementById("form");
form.onsubmit = e => {
  e.preventDefault();
  const name = nameInput.value = document.getElementById("name").value;
  const email = document.getElementById("email").value;
  const dob = document.getElementById("dob").value;
  const phone = document.getElementById("phone").value;
  const dept = document.getElementById("dept").value;
  if (phone.length !== 10 || isNaN(phone)) {
    alert("Invalid phone number");
    return;
  }
  document.getElementById("msg").innerText = "Registration Successful!";
  const fields = {
    ID: Math.floor(Math.random()*1000),
    Name: name,
    Email: email,
    DOB: dob,
    Dept: dept,
    Phone: phone
  };
  const tbody = document.getElementById("data");
  tbody.innerHTML = "";
  for (let key in fields) {
    tbody.innerHTML += `<tr><td>${key}</td><td>${fields[key]}</td></tr>`;
  }
}
```

```

    }
    document.getElementById("table").style.display = "table";
};
form.onreset = () => {
    document.getElementById("msg").innerText = "";
    document.getElementById("table").style.display = "none";
};
</script>
</body>
</html>

```

SQL

STEP 1: Create and select the database

```

CREATE DATABASE IF NOT EXISTS college_db;
USE college_db;

```

STEP 2: Create the students table

```

CREATE TABLE IF NOT EXISTS students (
    student_id INT AUTO_INCREMENT PRIMARY KEY,
    full_name VARCHAR(100) NOT NULL,
    email VARCHAR(100) NOT NULL UNIQUE,
    dob DATE NOT NULL,
    department VARCHAR(80) NOT NULL,
    phone CHAR(10) NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

```

STEP 3: Insert sample student records

```

INSERT INTO students (full_name, email, dob, department, phone) VALUES
    ('Arun Kumar', 'arun@student.edu', '2003-05-14', 'Computer Science',
'9876543210'),
    ('Priya Sharma', 'priya@student.edu', '2002-11-22', 'Electronics and Communication',
'8765432109'),
    ('Ravi Menon', 'ravi@student.edu', '2003-08-01', 'Mechanical Engineering',
'7654321098'),

```

('Sneha Iyer', 'sneha@student.edu', '2004-01-30', 'Electrical Engineering', '6543210987'),

('Karthik Raja', 'karthik@student.edu', '2002-07-19', 'Computer Science', '9123456780'),

('Divya Nair', 'divya@student.edu', '2003-03-11', 'Information Technology', '9988776655'),

('Mohammed Arif', 'arif@student.edu', '2002-09-05', 'Civil Engineering', '9871234560');

STEP 4: SELECT Queries - Retrieve Data

4a. Retrieve ALL student records

```
SELECT * FROM students;
```

4b. Retrieve specific columns only

```
SELECT student_id, full_name, email, department
```

```
FROM students;
```

4c. Filter by department

```
SELECT full_name, email, phone
```

```
FROM students
```

```
WHERE department = 'Computer Science';
```

4d. Sort students by name (A to Z)

```
SELECT full_name, department, dob
```

```
FROM students
```

```
ORDER BY full_name ASC;
```

4e. Count total students per department

```
SELECT department, COUNT(*) AS total_students
```

```
FROM students
```

```
GROUP BY department
```

```
ORDER BY total_students DESC;
```

4f. Find students born after 2003-01-01

```
SELECT full_name, dob, department
```

```
FROM students
```

```
WHERE dob > '2003-01-01'
```

```
ORDER BY dob;
```

4g. Search student by email

```
SELECT * FROM students
```

```
WHERE email = 'arun@student.edu';
```

4h. Limit output to first 3 records

```
SELECT * FROM students LIMIT 3;
```

STEP 5: UPDATE a record

```
UPDATE students
```

```
SET phone = '9000011111'
```

```
WHERE email = 'arun@student.edu';
```

Verify update

```
SELECT full_name, phone FROM students WHERE email = 'arun@student.edu';
```

STEP 6: DELETE a record

```
DELETE FROM students WHERE student_id = 7;
```

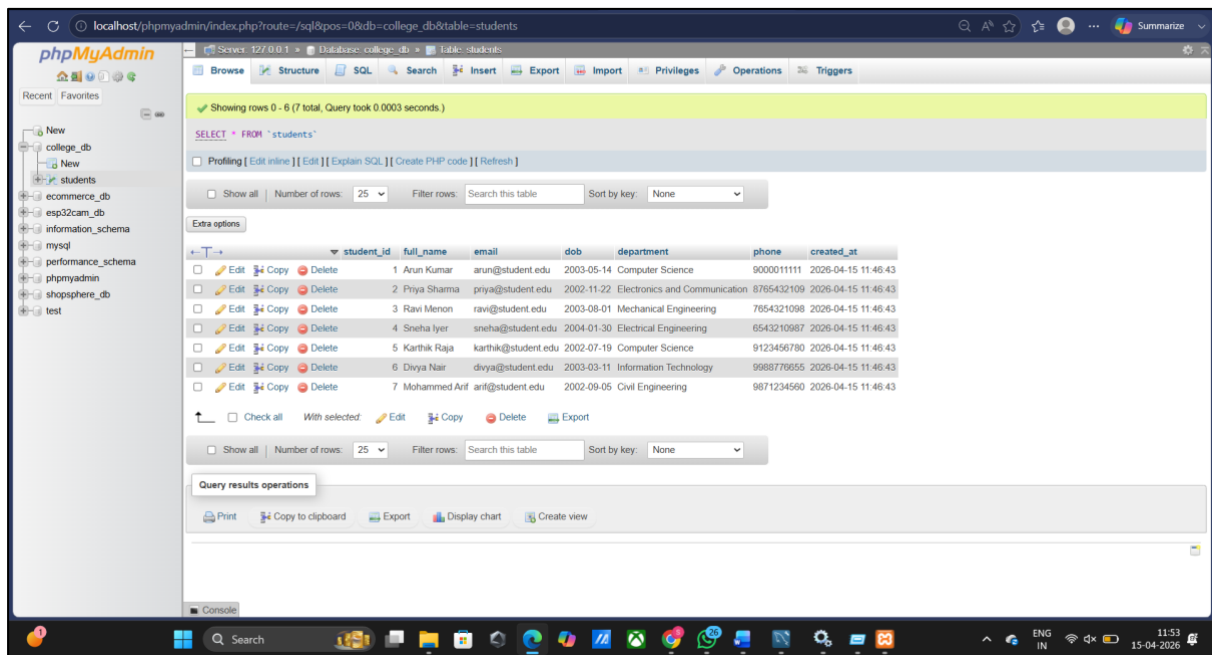
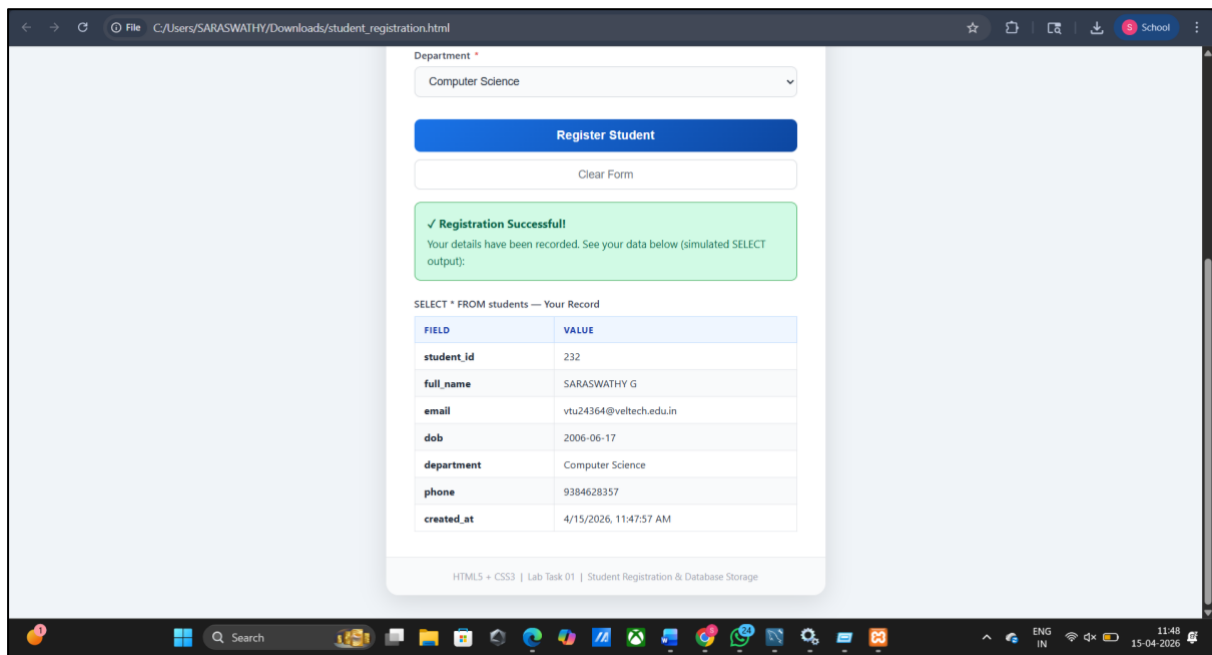
(Commented out to preserve data — uncomment to run)

STEP 7: Describe table structure

```
DESCRIBE students;
```

OUTPUT:

The screenshot shows a web browser window with the address bar displaying 'C:/Users/SARASWATHY/Downloads/student_registration.html'. The main content is a 'Student Registration' form. The form has a blue header with the title 'Student Registration' and a subtitle 'Fill in all required fields to register as a student.' The form fields are: Full Name (SARASWATHY G), Email Address (vtu24364@veltech.edu.in), Date of Birth (17-06-2006), Phone Number (9384628357), and Department (Computer Science). There is a 'Register Student' button and a 'Clear Form' button. The footer of the form says 'HTML5 + CSS3 | Lab Task 01 | Student Registration & Database Storage'.



RESULT:

The Student Registration Form was created using HTML5 and CSS3 with proper validation. A MySQL table students was designed with fields like student_id, full_name, email, dob, department, phone, and created_at. Data was inserted using INSERT and retrieved using SELECT, with filtering and grouping done using WHERE and GROUP BY, confirming successful database operations.